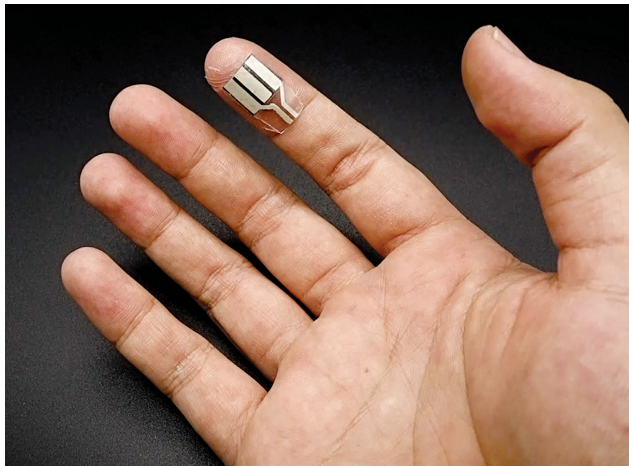


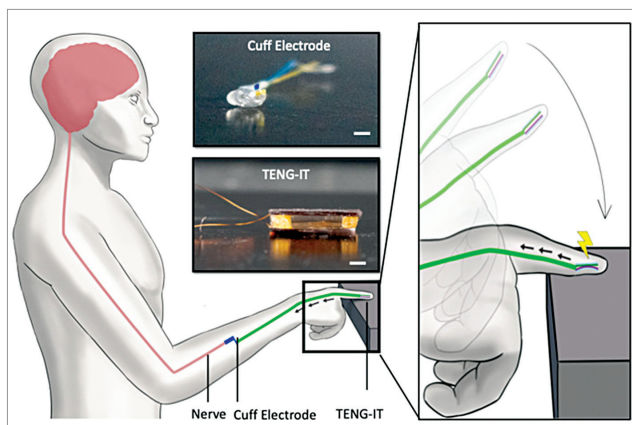
Fingertip sweat powered electronic device

Engineers at the University of California, San Diego, have developed a thin, flexible strip that can be worn on a fingertip to generate small amounts of electricity when a person's finger sweats or presses on it. It even generates power while the wearer is asleep, sitting still or doing activities involving light finger presses such as typing, texting, playing the piano, or tapping in Morse code. The device is made up of a padding of carbon foam electrodes that absorb sweat and convert it into electrical energy. These electrodes are equipped with enzymes that trigger chemical reactions between lactate and oxygen molecules in sweat to generate electricity. As the wearer sweats or presses on the strip, the electrical energy gets stored in a small capacitor and is discharged to other devices when needed.

This thin, flexible strip can be worn on a fingertip to generate small amounts of electricity when a person's finger sweats or presses on it (Credit: <https://ucsdnews.ucsd.edu>)



Technology restores sense of touch in damaged nerves

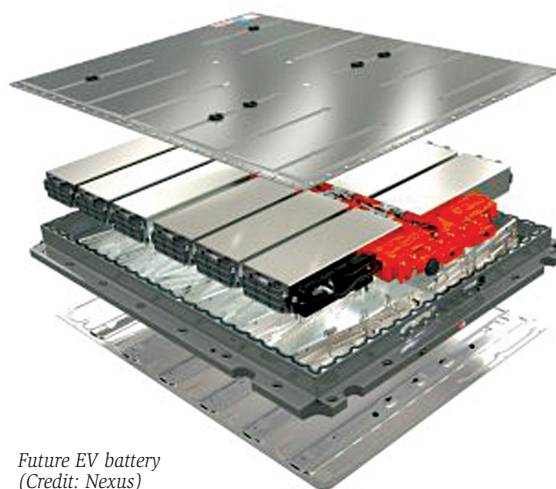


Restoring tactile sensation using a triboelectric nanogenerator (Credit: <https://pubs.acs.org>)

Researchers at Tel Aviv University have developed a sensor that can be implanted anywhere in the body, for example, under the tip of a severed finger. The sensor that is implanted in the nerve of the injured limb is connected directly to a healthy nerve. Each time the limb touches an object, the sensor is activated and conducts an electric current to the functioning nerve, which recreates the feeling of touch. The sensor does not require an external power source such as electricity or batteries. Instead, whenever the device senses friction, it charges itself. The device consists of two tiny plates, which release an electric charge when in contact with each other. This electric charge gets transmitted to the undamaged nerve. When the injured finger touches something, the touch releases tension corresponding to the pressure applied to the device.

Biodegradable EV battery from crop waste

Bhubaneswar based startup Nexus Power has developed a lithium-free biodegradable battery cell made from proteins derived from crop waste. The team leveraged the conductive properties of proteins to store and release electrical energy. Both electrodes and the electrolyte are made from bio-organic materials, and the aluminium casing is recyclable. Tests showed that the charging speeds for the green battery could be on par with a lithium battery of a similar capacity. It could even be faster. The team believes the technology will be fully commercialised in the next 2-3 years, and India could be the epicentre of the technology.



Future EV battery (Credit: Nexus)